2.2 Real Number System

First we shall recall how the real number system was developed. We start with natural numbers $\mathbb{N}$.

2.2.1 Rational Numbers

Note that $\mathbb{N} = \{1, 2, 3, \dots\}$ is enough for counting objects. In order to deal with loss or debts, we enlarged $\mathbb{N}$ to the set of all integers, $\mathbb{Z} = \{\dots, -4, -3, -2, -1, 0, 1, 2, \dots\}$, which consists of the natural numbers, zero, and the negatives of natural numbers. We call $\{0, 1, 2, 3, \cdots\}$ as the set of whole numbers and denote it by $\mathbb{W}$. Note that it differs from $\mathbb{N}$ by just one element, namely, zero. Now imagine dividing a cake into five equal parts, which is equivalent to finding a solution of $5x = 1$. But this equation cannot be solved within $\mathbb{Z}$. Hence we have enlarged $\mathbb{Z}$ to the set $\mathbb{Q} = \left\{ \frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0 \right\}$ of *ratios*; so we call each $x \in \mathbb{Q}$ as a *rational* number. Some examples of rational numbers are

$$-5, \ \frac{-7}{3}, \ \ 0, \ \frac{22}{7}, \ 7, \ 12.$$

We have seen in earlier classes that rational numbers are precisely the set of terminating or infinite periodic decimals. For example,

$$-5.0, \ -2.333\cdots, \ \frac{25}{99} = 0.252525\cdots, \ \frac{2}{3} = 0.66666\cdots, 7.14527836231231231\cdots$$

are rational numbers.

2.2.2 The Number Line

Let us recall "*The Number Line*". It is a horizontal line with the *origin*, to represent 0, and another point marked to the right of 0 to represent 1. The distance from 0 to 1 defines one unit of length. Now put 2 one unit to the right of 1. Similarly we put any positive rational number x to the right of 0 and x units away. Also, we put a negative rational number $-r, r > 0$, to the left of 0 by r units. Note that for any $x, y \in \mathbb{Q}$ if $x < y$, then x is to the left of y; also $x < \frac{x+y}{2} < y$ and hence between any two distinct rational numbers there is another rational number between them.

Question:

Have we filled the whole line with rational numbers?

The answer to the above question is "No" as the following consideration demonstrates. Consider a square whose side has length 1 unit. Then by Pythagoras theorem its diagonal has length $\sqrt{2}$ units.

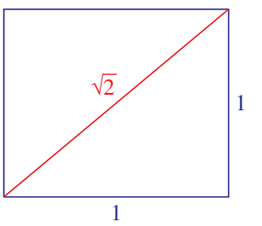


Figure 2.1